00212



Sri Lanka Institute of Information Technology

B.Sc. Honours Degree in Information Technology

Specialized in Information Technology

Final Examination
Year 3, Semester 2 (2022)

IT3060 – Human-Computer Interaction

Duration: 2 Hours

November 2022

Instructions to Candidates:

- This paper has 4 questions.
- Answer all questions in the booklet given.
- The total marks for the paper is 100.
- This paper contains 6 pages, including the cover page.
- Electronic devices capable of storing and retrieving text, including calculators and mobile phones are not allowed.
- 10 minutes of reading time is required.

Question 1 **(25 Marks)**

a. **"User experience is contextual".** Briefly explain its application with a suitable example.

(06 marks)

b. Assume that you have decided to help Sri Lankan people who are having hard time to source medicine and medical supplies around the area. To tackle that problem, your idea is to create a platform called "**Medicineneeds.lk**" to request and search for medicine and medical supplies for the people, who are unable to source medicine due to the current situation in the country.

I. The design-thinking model will be useful for your design process. Name the phases in the design-thinking model and indicate one (01) important task in each phase.

(10 marks)

II. Identify 4 user groups of **Medicineneeds.lk**.

(04 marks)

c. Ideation is at the heart of the Design Thinking process. There are hundreds of ideation techniques. "Brainstorm" is one main popular techniques for Ideation. How can Brainstrom technique can be applied to generate ideas to solve the above problem?

(05 marks)

Question 2 **(25 Marks)**

a. **'Prototyping is an integral part of the design process'** State whether you "agree" or "disagree" with the given statement. Justify your answer.

(04 Marks)

b. Compare and contract high-fidelity and low fidelity of prototypes referring to Content, Interaction, and aesthetics.

(05 Marks)

c. A Leading Vehicle Manufacturing company is planning to release a new **Smart Concept Car** *(for domestic use)* during the Geneva Motor Show in 2024. The new concept car is expected to be controlled with a fully digitalized automobile control system called **"ADAS: Advanced Driver Assistance System"**.

You have been selected as one of the **Automobile Control Software** designers for the new car. The new Futuristic car design is required to be highly focused on the following **technological leaps** in the automotive industry.

1. **Connected Car Technology**
   A modern vehicle is equipped with lots of sensors, gauges, and microchips. Connected car technologies allow for timelier vehicle maintenance, fewer malfunctions, and more data for further analysis. *(Eg: Tyre Pressure, Road Condition, Passenger Health Monitoring, Suspension Adjust, Vehicle Ambient control (Light and Climate), Entertainment System and Virtual Mechanic for Self-diagnose and Maintenance, etc.)*

2. **Automotive Cloud Solutions**
   Updates over wireless networks just like a regular smartphone, enabling owners to download the latest software version anytime-anywhere. *(Over the Air: OTA Update)*. This will keep the vehicle up-to-date with latest software, enhancing its capabilities with the latest features.

3. **Autonomous Driving**
   When computers start controlling vehicle movements, many categories of people who cannot drive now because of restrictions *(senior citizens, people with disabilities, children)* will break free of this limitation. *(Auto Lane Departure, Collision Control, Auto Parking, Cruise Control, Auto Summoning, etc.)*

4. **Mobility as a Service (MaaS)**
   This is an emerging type of service that, through a joint digital channel enables users to plan, book, and pay for multiple types traveling by **shifting away from the traditional concept of personally owned vehicles** and towards mobility provided as a service, which users can pay for with a single account.

Before going into the initial software design phase, you have to conduct user research by interviewing as many users as necessary.

I. Identify three stakeholders of this system '**ADAS**'

(03 marks)

II. Construct a list of questions that can be used in a **semi-structured interview** based on the above-mentioned technological factors. You are expected to prepare 10 questions.

(05 marks)

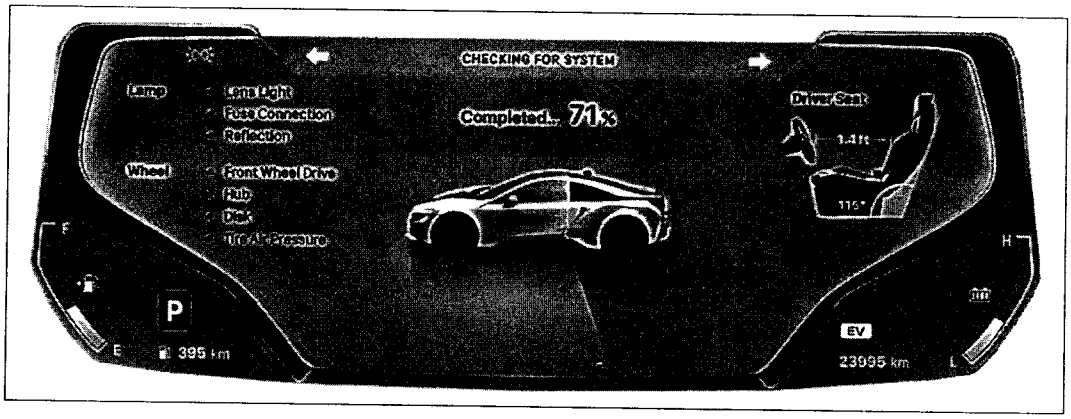
Hint: *Use the pointers given in the description effectively.*

III. Carefully study the given **High-Fidelity UI** concept prototype samples (Sample 1 and Sample 2), representing vehicle functions, designed for the **ADAS**.

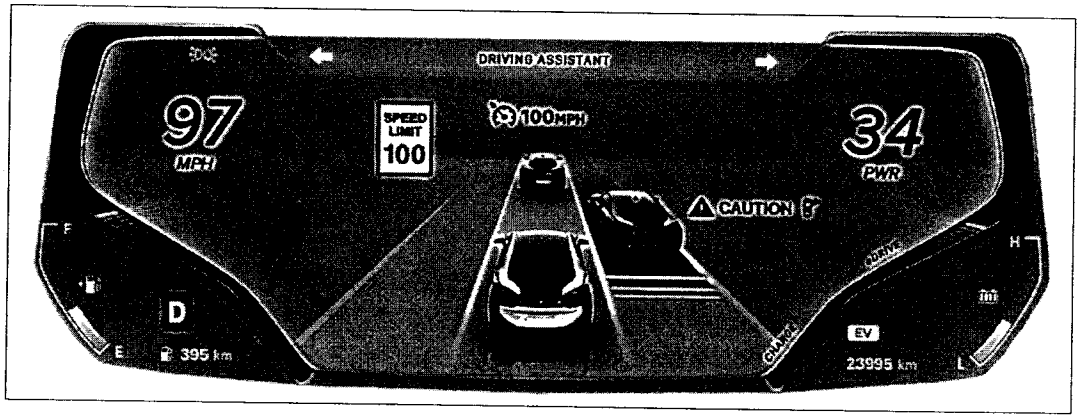
Analyze and identify how effectively following (i-iv) **UI design principles** have been used in the concept design. Choose a UI element from the given samples and elaborate *(Eg: Display of Speedometer)*.

(08 marks)

i. Visibility
ii. Feedback
iii. Affordance
iv. Mapping



Sample 1



Sample 2

Question 3 **(25 Marks)**

a. Heuristic Evaluation is a proper technique for measuring the usability of an interface. What do you mean by Heuristic Evaluation?

(04 Marks)

b. The following UI image shows an error message displayed in windows heuristic evaluation principles has to be considered to improve the user interface (Image 1) shown below.

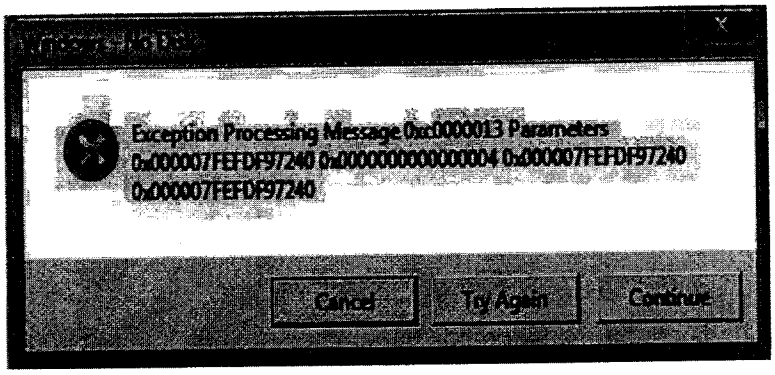


Image 1

(06 Marks)

c. Assume that you are given the task of designing an interactive Kiosk for a Museum to be used by school children. The kiosk will provide learning activities and games about the water cycle.

I. Choose a typical user profile, and produce a detailed, complete persona.

(08 Marks)

II. Formulate a detailed main scenario for the selected user above.

(07 Marks)

Question 4 **(25 Marks)**

a. Peoples hospital has decided to redesign its Medicare application after getting many complaints from users about the application. The last time when the application was designed there was no participation of stakeholders in the designing process.

What should be the best technique that would involve stakeholders in the design process?

(03 marks)

b. Explain what the **'Think Aloud Evaluation Technique'** is and justify why it's important using examples.

(08 marks)

c. Interactions are 'what the user is doing when interacting with a system', Briefly describe the four types of interactions used in the process of implementation of a user interface.

(08 marks)

d. compare formative and summative evaluation. When would be used in the design process?

(06 marks)

========== End of the Question Paper ==========